

ASTR211 Data Reduction Log

Date: 15/8/24

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Dataset Option: Spectroscopy

Process your data using code from previous Labs.

Fill in the below sections as you work through the lab.

Answer the questions below and also note any problems you encounter.

For each problem document:

1. What is the problem?
 2. How did you find it?
 3. How does this affect the raw, calibration and/or science images?
 4. How did/would you solve it?
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1. Summarise of your data (Task 1)

How many of each type of image file.

- 5 darks
- 1 ThAr
- 10 whites
- 1 Raw

Do any of the images look like they have any errors. Discuss with tutor.

Raw, Darks look good.

One of the whites is bad, popping it from list.

ThAr is good.

Was having minor problem with slow display of images. Resolved by restarting.

Sorted.

2. Calibration files (Task 2)

Comment on your calibration files compared to the individual files.

Master dark came out good

Having an issue with master white, np.median seems to not be working. I was appending to the wrong list. Solved.

Median master white looks good

Mean has a clear shadow where one of the whites has been shifted down. We will use the median.

3. Reducing and order tracing (Task 3)

Comment on how you improved the order tracing. How many orders did you find?

There are 7 orders, It was pretty straightforward. Set the height of `find_peaks()` to 2000 to clear any minor bumps at the bottom of the orders.

4. Extracting one order (Task 4)

Which order did you choose to extract?

Chose to go for the H-alpha in the 5th order

Discuss why H-alpha is in multiple orders.

It's an artifact of how the diffraction grating works. The same wavelength can appear in multiple orders at different angles. $d \sin \theta = n \lambda$ has multiple solutions.

5. Navigating a ThAr Atlas (Task 5)

Describe the contents of the ThAr linelist. How did you modify the linelist? Describe your process for extracting the relevant information. Linelist contained columns of wavelength, relative strength, and two sets of orders.

Used `np.loadtxt`, skipping the first row. Then iterated over each line checking `line[2] == 5` or `line[3] == 5` and appending relevant candidates to a new list.

6. Mapping pixels to wavelengths I (Task 6)

How many thorium lines are in your order? Discuss how you find the pixel values for your lines. What to do the height and distance arguments control?

5 lines in the order. Height control the minimum height for a line to count. Distance is the minimum horizontal distance between peaks. Finding the pixel values is fairly simple. just used `find_peaks()` again.

7. Mapping pixels to wavelengths II (Task 7)

Describe the pixel-wavelength space. What does the relation look like? What polynomial did you use to fit the relation?

Relation is linear. Used a 1st order polynomial to fit relationship. Was having trouble originally. Realised I needed to sort the lambda values to see the linear relationship.

8. Quality Checks (Task 8)

Based on the quality check, is your wavelength calibration correct? Discuss.

Yes. It lines up perfectly. Had to scale flux by about 2500

9. Wavelength calibrate all orders (Task 9)

Discuss any issues encountered while wavelength calibrating the remaining orders.
